

Yehoshua ("Shua") Halle

yehoshua.halle@outlook.com | 301-557-0573 | Portfolio: shua5115.github.io

EDUCATION

Carnegie Mellon University (CMU) Pittsburgh, PA

May 2027

Master of Robotics Systems Development | GPA: 3.67/4.00

Relevant courses: Autonomous Vehicles; Robot Autonomy; Robot Mobility; Computer Vision; Robotics Business

University of Maryland (UMD) College Park, MD

December 2024

Bachelor of Mechanical Engineering | GPA: 3.84/4.00 | Dean's List Honors

Minor of Robotics and Autonomous Systems

PROJECTS

MRSD Capstone, Automated Lab Robot, "Catalyst", CMU (Pittsburgh, PA)

Aug 2025 - Dec 2026

- Designed a mobile manipulator system to automate experiments by moving well plates between instruments
- Integrated 3D SLAM library into ROS 2 by writing interfacing code, achieving 10cm-accurate localization
- Implemented force-feedback admittance control on XArm6 resulting in safe, accurate, and repeatable pick/place
- Discovered the root cause of critical networking issues through unit testing, ROS debugging, Linux sysadmin tools, and maintaining docs. Resulted in improved system reliability and timely completion of milestones

F1Tenth Autonomous Racing, CMU (Pittsburgh, PA)

Jan 2026 - May 2026

- Developed autonomous driving software for an RC car with NVidia Jetson compute
- Programmed RRT with Pure Pursuit to follow an optimal racing line with collision avoidance, applied in race to plan around obstacles designed to trick simpler methods
- Achieved 1st place in time trial races between 12 teams by optimizing reliability and speed in physical tests

Powertrain Team Leader, Terps Racing Baja SAE Team (College Park, MD)

Jan 2023 - Aug 2024

- Coordinated a team of 10 students to design and manufacture a custom powertrain for an off-road vehicle
- Optimized transmission using C++ simulation and physical trials to improve 150 ft acceleration time by 1.6s: from 5.98s (top 25%) to 4.34s (top 15%)
- Designed a belt-driven 4WD system using physically-derived loads with FEA. System had 100% reliability during 4 hour endurance race. My FEA procedure was used as an example for teaching new members

"Assistive Feeder" Project, Volunteers for Medical Engineering (College Park, MD)

Jan 2023 - May 2024

- Designed a robot to help people with disabilities eat meals with more independence at an affordable price
- Programmed kinematics solver with current-sensing collision detection for Arduino microcontroller in C++, allowing the robot to be configurable for different users' needs and to react to collisions safely
- Tested prototype with end user with cerebral palsy. Received positive feedback on food serving function and criticisms to improve the user experience. I refined the design to address feedback

WORK EXPERIENCE

Program Director of Computer Science, TIC Summer Camp (Bethesda, MD)

July 2025 - Aug 2025

- Managed a team of 10 counselors to teach programming, robotics, and web concepts to kids of ages 6-15
- Trained counselors on technical communication by applying the socratic method and psychology. The CEO recognized its positive impact and incorporated my training material at all other camp locations
- Resolved camper misbehavior and counselor conflict by approaching calmly, understanding each perspective, collaboratively creating a plan to improve, and holding accountable to the plan, leading to few repeat offenses

SKILLS

Software: ROS/ROS 2, MATLAB, Simulink, SOLIDWORKS, Inventor, Fusion 360, Linux OS, Microsoft Office

Programming Languages: Python, C++, C, CMake, Java, Lua, Bash

Engineering: CAD, CAM, GD&T, FEA, DFM, DFA, Physics Simulation, 3D Printing, Milling, Lathe Turning

Math: Controls, Optimization Differential Equations, Linear Algebra, Statistics